

QUAID-I-AZAM UNIVERSITY ISLAMABAD
ASSIGNMENT #2



Subject: Cloud DevOps

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Task 1: Create a Docker Image with Git Installed

In this task, we created a Docker image that includes Git using Alpine as the base image.

Steps performed:

1. Created a Dockerfile with the following contents:

```
FROM alpine:latest
RUN apk update && apk add git
CMD ["git", "--version"]
```

2. Built the image using:

```
docker build -t qu1:v1.0 .
```

3. Verified the image using:

```
docker images
```

4. Created a container and ran it in the background:

```
docker run -d --name git_container qu1:v1.0
```

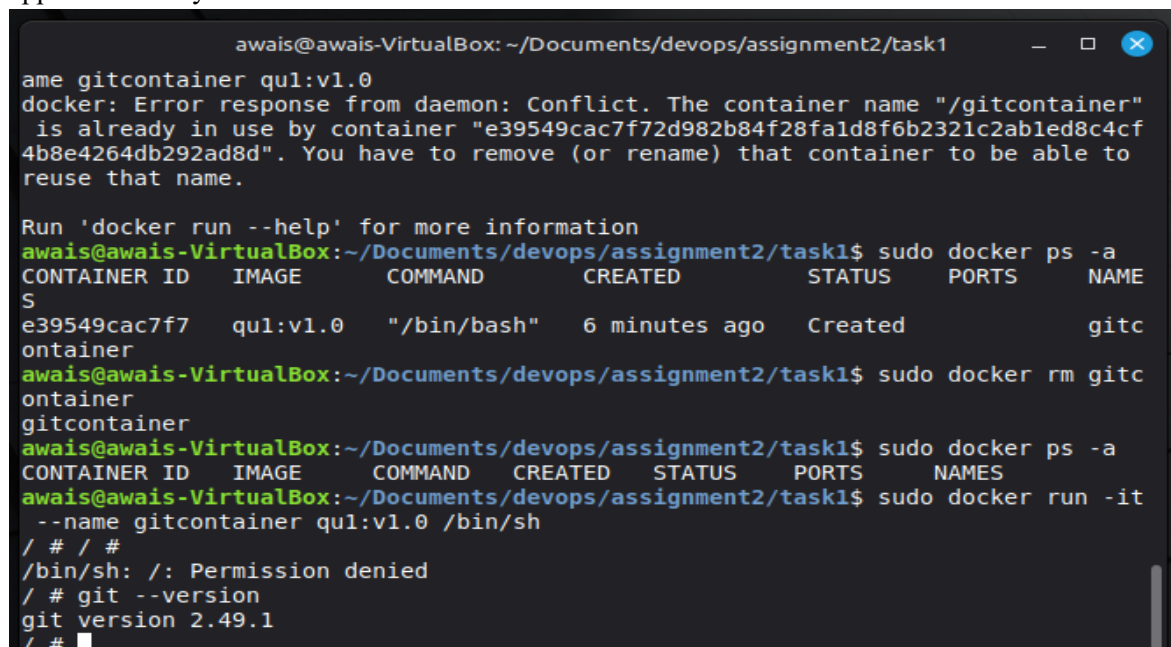
5. Attached to the container using:

```
docker attach git_container
```

6. Verified that Git was installed inside the container using:

```
git --version
```

Observation: The image was successfully created with Git installed, and the tag `qu1:v1.0` was applied correctly.



```
awais@awais-VirtualBox: ~/Documents/devops/assignment2/task1
ame gitcontainer qu1:v1.0
docker: Error response from daemon: Conflict. The container name "/gitcontainer"
is already in use by container "e39549cac7f72d982b84f28fa1d8f6b2321c2abled8c4cf
4b8e4264db292ad8d". You have to remove (or rename) that container to be able to
reuse that name.

Run 'docker run --help' for more information
awais@awais-VirtualBox:~/Documents/devops/assignment2/task1$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
e39549cac7f7   qu1:v1.0  "/bin/bash"             6 minutes ago Created                        gitc
ontainer
awais@awais-VirtualBox:~/Documents/devops/assignment2/task1$ sudo docker rm gitc
ontainer
gitcontainer
awais@awais-VirtualBox:~/Documents/devops/assignment2/task1$ sudo docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
awais@awais-VirtualBox:~/Documents/devops/assignment2/task1$ sudo docker run -it
--name gitcontainer qu1:v1.0 /bin/sh
/ # / #
/bin/sh: /: Permission denied
/ # git --version
git version 2.49.1
/ #
```

Note: Faced many errors during the process. The final screenshot for Task 1 was taken after successfully completing the task. All the commands mentioned in the report were used, and the screenshot was captured once the task was completed.

Task 2: Working with Docker Images and Containers

In this task, we explored Docker image management and container operations.

Steps performed:

1. Modify the Docker file:

```
FROM python:3.9-alpine
ADD . /code
WORKDIR /code
RUN pip install -r requirements.txt
```

2. Now create another docker-compose.yml file

```
version: "3.8"
services:
  web:
    build: .
    command: python app.py
    ports:
      - "5000:5000"
    volumes:
      - code_volume:/code
    networks:
      - app-network
networks:
  app-network:
    driver: bridge
volumes:
  code_volume:
```

3. Now Build the app and run it using this command:

```
docker compose up -d --build
```

4. Listed all running containers:

```
docker ps
```

5. Verify the app:

<http://localhost:5000>

6. View created network and volume:

```
docker network ls
```

```
docker volume ls
```

Use the another terminal to view network, keep app running while checking network and volume in other terminal

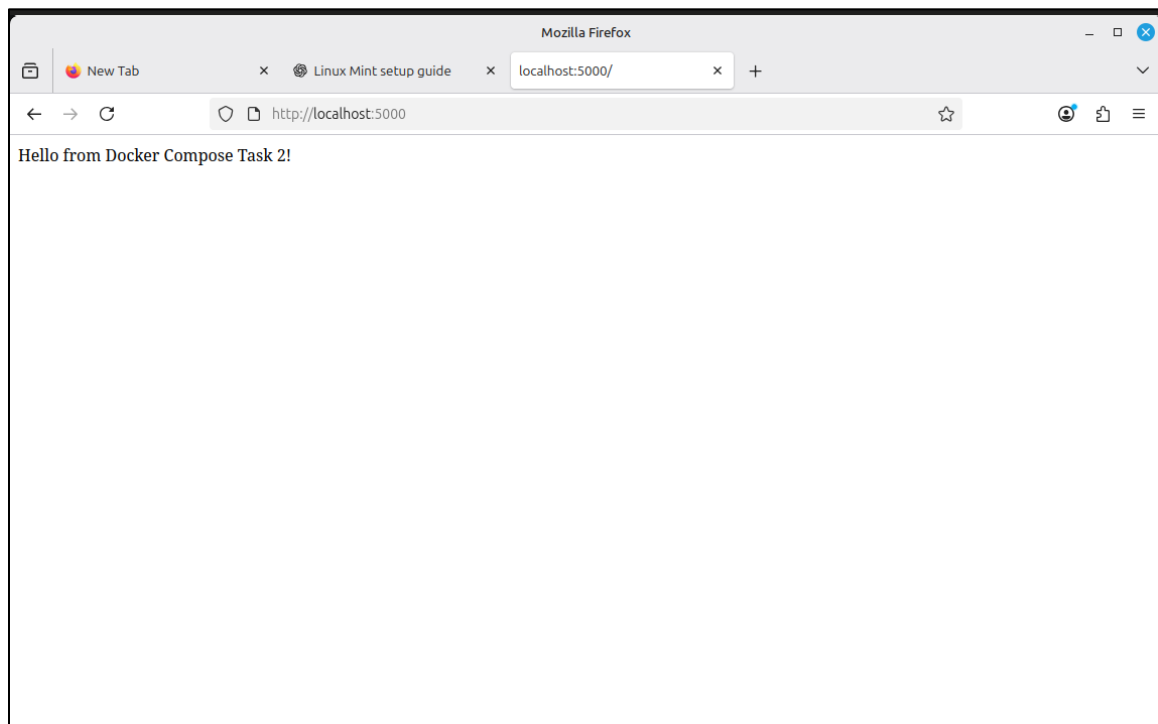
```
awais@awais-VirtualBox: ~/Documents/devops/assignment2/task2

volumes:
  code_volume:

awais@awais-VirtualBox:~/Documents/devops/assignment2/task2$ sudo docker compose up --build
WARN[0000] /home/awais/Documents/devops/assignment2/task2/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 13.9s (11/11) FINISHED
=> [internal] load local bake definitions
=> => reading from stdin 529B
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 246B
=> [internal] load metadata for docker.io/library/python:3.9-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 419B
=> CACHED [1/4] FROM docker.io/library/python:3.9-alpine@sha256:f4b621
=> [2/4] ADD . /code
=> [3/4] WORKDIR /code
=> [4/4] RUN pip install -r requirements.txt
=> exporting layers
=> writing image sha256:9b478aldeca7055a4f34db7cee2e77c4577fecbc
=> naming to docker.io/library/task2-web
=> resolving provenance for metadata file
[+] Running 4/4
✔ task2-web Built
✔ Network task2_app-network Created
✔ Volume task2_code_volume Created
✔ Container task2-web-1 Recreated
Attaching to web-1
web-1 | * Serving Flask app 'app'
web-1 | * Debug mode: off
web-1 | WARNING: This is a development server. Do not use it in a production
web-1 | * Running on all addresses (0.0.0.0)
web-1 | * Running on http://127.0.0.1:5000
web-1 | * Running on http://172.19.0.2:5000
web-1 | Press CTRL+C to quit

[+] Enable Watch
```

```
awais@awais-VirtualBox: ~
awais@awais-VirtualBox:~$ sudo docker network ls
[sudo] password for awais:
Sorry, try again.
[sudo] password for awais:
NETWORK ID      NAME      DRIVER      SCOPE
27b6ef4541a8    bridge    bridge       local
12eb8eef0045    host      host         local
84431daabdd6b   none      null         local
7abeeeab464e2   task2_app-network    bridge       local
89d06542ff03    task2_mynetwork      bridge       local
awais@awais-VirtualBox:~$ sudo docker volume ls
DRIVER      VOLUME NAME
local       task2_code_volume
awais@awais-VirtualBox:~$
```



Observation: This task demonstrated the use of Docker commands to manage containers and images effectively.

Task 3: Running a Web Application using Docker Compose

In this task, we configured a three-service web application using Docker Compose: a PHP-Apache web server, a MySQL database, and phpMyAdmin for database management.

Steps performed:

1. Created a `docker-compose.yml` file with the following configuration:
version: "3.9"

services:

db:

```
image: mysql:latest
container_name: mysql_db
restart: always
ports:
  - "9906:3306"
environment:
  MYSQL_ROOT_PASSWORD: root
  MYSQL_DATABASE: studentdb
  MYSQL_USER: user
  MYSQL_PASSWORD: user123
volumes:
  - db_data:/var/lib/mysql
```

php-apache-environment:

```
container_name: php-apache
build:
  context: .
  dockerfile: Dockerfile
restart: always
depends_on:
  - db
ports:
  - "8000:80"
volumes:
  - ../var/www/html/
environment:
  MYSQL_HOST: db
  MYSQL_USER: user
  MYSQL_PASSWORD: user123
  MYSQL_DATABASE: studentdb
```

phpmyadmin:

```
image: phpmyadmin:latest
container_name: phpmyadmin_ui
restart: always
depends_on:
  - db
ports:
  - "8081:80"
```

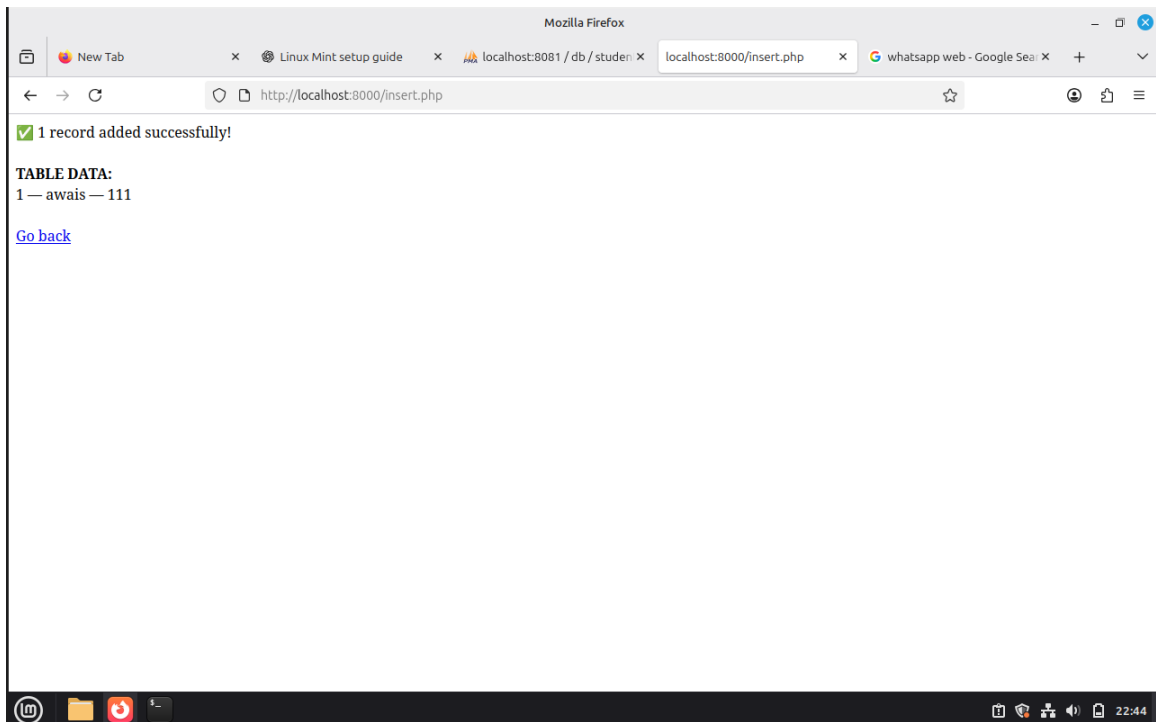
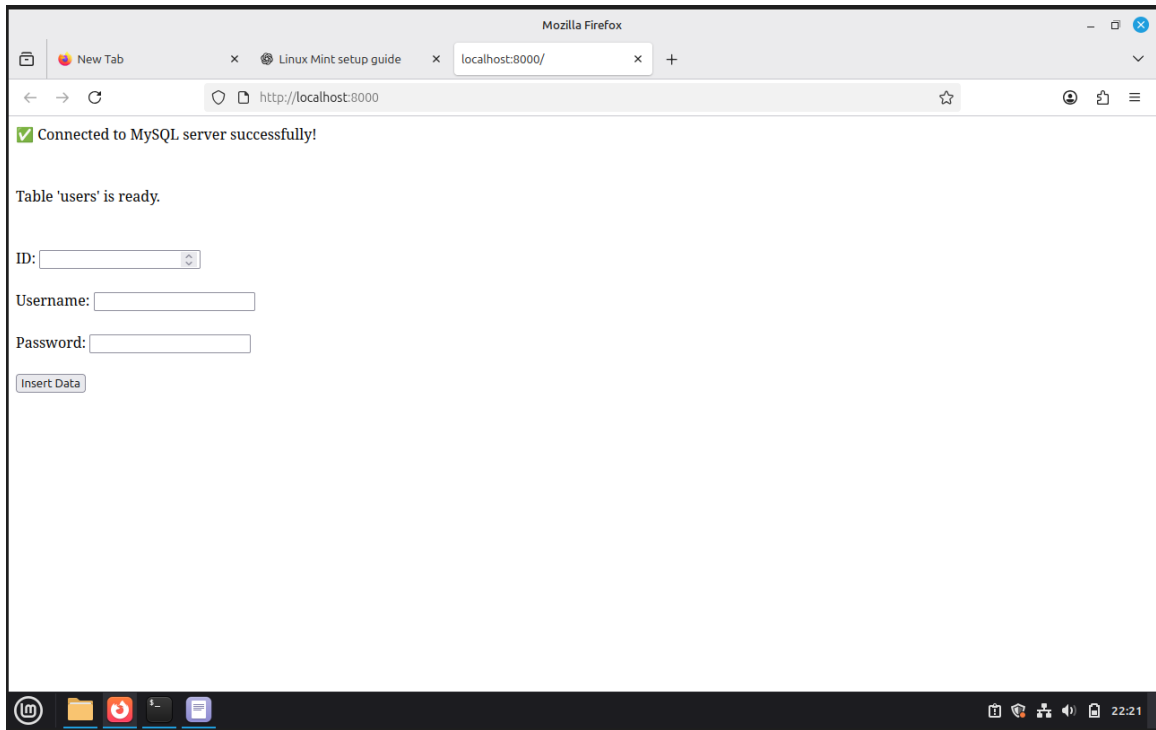
```
environment:
  PMA_HOST: db
  PMA_USER: root
  PMA_PASSWORD: root
```

```
volumes:
  db_data:
```

2. Started the containers using:
`docker compose up -d --build`
3. Verified containers were running:
`docker ps`
4. Accessed the PHP application via `http://localhost:8080`.
5. To end the container use these command:
`docker compose down`
`docker compose restart db`

Observation: Successfully deployed a PHP-MySQL web application using Docker Compose.

```
awais@awais-VirtualBox: ~/Documents/devops/assignment2/task3
File Edit View Search Terminal Help
✓ Container phpmyadmin ui Removed 2.1s
✓ Container php-apache Removed 2.3s
✓ Container mysql_db Removed 1.7s
✓ Volume task3 db_data Removed 0.0s
✓ Network task3 default Removed 0.4s
WARN[0000] /home/awais/Documents/devops/assignment2/task3/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 1.8s (9/9) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 605B 0.0s
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 163B 0.1s
=> [internal] load metadata for docker.io/library/php:8.0-apache 1.2s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/php:8.0-apache@sha256:c87b4c94d53b355204 0.0s
=> CACHED [2/3] RUN docker-php-ext-install mysqli && docker-php-ext-enab 0.0s
=> CACHED [3/3] RUN apt-get update && apt-get upgrade -y 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:56bf6441ddf05ace7f04d9fbeb3db6fd4d227c8adbbe8 0.0s
=> => naming to docker.io/library/task3-php-apache-environment 0.0s
=> => resolving provenance for metadata file 0.0s
[+] Running 6/6
✓ task3-php-apache-environment Built 0.0s
✓ Network task3 default Created 0.2s
✓ Volume task3 db_data Created 0.0s
✓ Container mysql_db Started 1.2s
✓ Container phpmyadmin ui Started 3.5s
✓ Container php-apache Started 3.0s
awais@awais-VirtualBox:~/Documents/devops/assignment2/task3$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
6f4f5a8e4223   task3-php-apache-environment        "docker-php-entrypoi..." About a minute ago Up About a minute 0.0.0.0:8000->80/tcp, [::]:8000->80/tcp
3a7045e6884e   phpmyadmin:latest                   "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
0d0bd962101d   mysql:latest                         "docker-entrypoint.s..." About a minute ago Up About a minute 33060/tcp, 0.0.0.0:9906->3306/tcp, [::]:9906->3306/tcp
```



localhost:8081 / db / studentdb / users | phpMyAdmin 5.2.3 — Mozilla Firefox

Server: db » Database: studentdb » Table: users

Showing rows 0 - 0 (1 total, Query took 0.0087 seconds.)

```
SELECT * FROM `users`
```

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☐ Show all | Number of rows: 25 | Filter rows:

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